

3.2 Deplecijska sila med ploščama v suspenziji trdih kroglic
↳ (hpr pesek)

$$\delta = 100 \text{ nm}$$

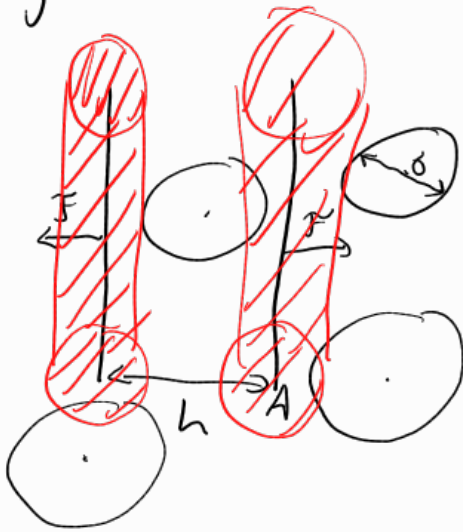
$$A = 1 \text{ mm}^2$$

$$h = 75 \text{ nm}$$

$$\frac{N}{V} = 10^{18} \text{ m}^{-3}$$

$$T = 20^\circ \text{C}$$

$$F(h) = ?$$



$$dF = -S d\Gamma + F dh$$

$$F = \frac{1}{\beta} \left(\frac{\partial \beta F}{\partial h} \right)_i \quad \text{ker } \beta \text{ ni odvisno od } h$$

$$e^{-\beta F} = c \left[\int_{V_{\text{razp. vol.}}} e^{-\beta \frac{\vec{p}^2}{2m}} d\vec{p} \int d\vec{s} \right]^N = c \left(\frac{2\pi m}{\beta} \right)^{3N/2} V_r^N(h)$$

$$1) h > \delta: V_r = V - 2A\delta$$

$$2) h < \delta: V_r = V - A(h + \delta)$$

$$-\beta F = \ln C + \frac{3N}{2} \ln \frac{2\pi m}{\beta} + N \ln V_r$$

$$F(h) = F_0(T) - N k_B T \left(\ln V + \underbrace{\ln \left(1 - \frac{A}{V} (h + \delta) \right)}_{\frac{A}{V} (h + \delta)} \right)$$

$$F = \left(\frac{\partial F}{\partial h} \right)_T = \frac{N}{V} A k_B T = \underline{\underline{4 \cdot 10^{-9} \text{ N}}}$$

3.8 Dvonivajski sistem

$$\varepsilon \text{ ————— } N_2 \quad S(N_2) = ?$$

$$0 \text{ ————— } N_1 \quad \langle N_2 \rangle(t) = ?$$

$$S(N_2) = k_B \ln \binom{N}{N_2} = k_B \ln \frac{N!}{N_2! (N-N_2)!} = \dots = -Nk_B \left[\frac{N_1}{N} \ln \frac{N_1}{N} + \frac{N_2}{N} \ln \frac{N_2}{N} \right]$$

$$F = \min$$

$$F(N_2) = E(N_2) - TS(N_2) = N_2 \varepsilon - Tk_B (N \ln N - N_2 \ln N_2 - (N-N_2) \ln (N-N_2))$$

$$\therefore \frac{d}{dN_2}$$

$$0 = \varepsilon - Tk_B \left(-N_2 \frac{1}{N_2} - \ln(N_2) + \ln(N-N_2) - (N-N_2) \frac{1}{N-N_2} \cdot (-1) \right)$$

$$\ln \frac{N-N_2}{N_2} = \frac{\varepsilon}{k_B T} \rightarrow \frac{N-N_2}{N_2} = e^{\frac{\varepsilon}{k_B T}} \rightarrow N_2 = N \left(\frac{1}{e^{\frac{\varepsilon}{k_B T}} + 1} \right)$$

$$p_2 = \frac{N_2}{N} = \frac{1}{e^{\frac{\varepsilon}{k_B T}} + 1} = \frac{e^{-\beta \varepsilon}}{1 + e^{-\beta \varepsilon}}$$

$$p_1 = 1 - p_2 = \frac{1}{1 + e^{-\beta \varepsilon}}$$